

Streamlit

1. Introduction to Streamlit

Streamlit is a Python library used to build **interactive web applications** using only Python. You don't need HTML, CSS, or JavaScript.

It is mainly used for:

- Data dashboards
- Machine learning model interfaces
- Data analysis and visualization tools

Installation

pip install streamlit

Run a Streamlit App

Save your Python file (for example app.py) and run:

```
streamlit run app.py
```

or

```
python -m streamlit run fun.py
```

2. Basic Structure of a Streamlit App

import streamlit as st

```
st.title("My First Streamlit App")
```

```
st.header("Welcome to Streamlit")
```

```
st.subheader("A simple web app using Python")
```

```
st.text("This is normal text")
```

```
st.markdown("**This is bold text using Markdown**")
```

Explanation:

- st.title() – Adds the main title
 - st.header() – Adds a header
 - st.subheader() – Adds a smaller title
 - st.text() – Displays plain text
 - st.markdown() – Supports markdown formatting
-

3. Displaying Data

Streamlit can display lists, dictionaries, or Pandas DataFrames directly.

```
import streamlit as st

import pandas as pd

data = {
    "Name": ["Atharva", "Riya", "Karan"],
    "Score": [85, 92, 78]
}

df = pd.DataFrame(data)
```

```
st.write("Displaying a DataFrame:")

st.dataframe(df) # interactive table

st.table(df)    # static table
```

4. User Input Widgets

Widgets allow user interaction with your app.

```
import streamlit as st

name = st.text_input("Enter your name:")

age = st.number_input("Enter your age:", min_value=1, max_value=100)

gender = st.radio("Select gender:", ("Male", "Female", "Other"))

hobbies = st.multiselect("Select hobbies:", ["Reading", "Music", "Sports", "Coding"])

rating = st.slider("Rate this app:", 0, 10, 5)

if st.button("Submit"):
    st.write(f"Name: {name}")
    st.write(f"Age: {age}")
    st.write(f"Gender: {gender}")
    st.write(f"Hobbies: {hobbies}")
    st.write(f"Rating: {rating}")
```

Explanation:

- `st.text_input()` – For text input
 - `st.number_input()` – For numeric input
 - `st.radio()` – Choose one option
 - `st.multiselect()` – Choose multiple options
 - `st.slider()` – Pick value from range
 - `st.button()` – Trigger an action
-

5. File Uploading

You can upload files like CSVs or images.

```
import streamlit as st
```

```
import pandas as pd
```

```
uploaded_file = st.file_uploader("Upload a CSV file", type=["csv"])
```

```
if uploaded_file is not None:
```

```
    df = pd.read_csv(uploaded_file)
```

```
    st.write("Preview of uploaded file:")
```

```
    st.dataframe(df.head())
```

6. Displaying Images, Audio, and Video

```
import streamlit as st
```

```
st.image("example.jpg", caption="Sample Image", width=300)
```

```
st.audio("sample.mp3")
```

```
st.video("sample.mp4")
```

7. Layout and Organization

You can arrange content in sidebars, columns, or tabs.

Sidebar

```
import streamlit as st
```

```
st.sidebar.title("Sidebar Example")

name = st.sidebar.text_input("Enter your name:")

st.write(f"Hello, {name}")
```

Columns

```
col1, col2 = st.columns(2)

col1.write("This is column 1")

col2.write("This is column 2")
```

Tabs

```
tab1, tab2 = st.tabs(["Home", "About"])

with tab1:

    st.write("Welcome to the Home tab")

with tab2:

    st.write("About this project...")
```

Expanders

```
with st.expander("See details"):

    st.write("This section is hidden until clicked.")
```

8. Forms and Submit Buttons

Forms group multiple input fields and submit them together.

```
import streamlit as st
```

```
with st.form("student_form"):

    name = st.text_input("Name")

    age = st.number_input("Age", min_value=1)

    submitted = st.form_submit_button("Submit")
```

```
if submitted:

    st.write(f"Student Name: {name}, Age: {age}")
```

9. Download Buttons

Allow users to download data as files.

```
import streamlit as st

import pandas as pd

df = pd.DataFrame({"Name": ["A", "B"], "Marks": [90, 80]})

csv = df.to_csv(index=False).encode("utf-8")

st.download_button(
    label="Download CSV",
    data=csv,
    file_name="data.csv",
    mime="text/csv"
)
```
